

Experiment No. 17

Write a Prolog Program for a DB with Name, DOB

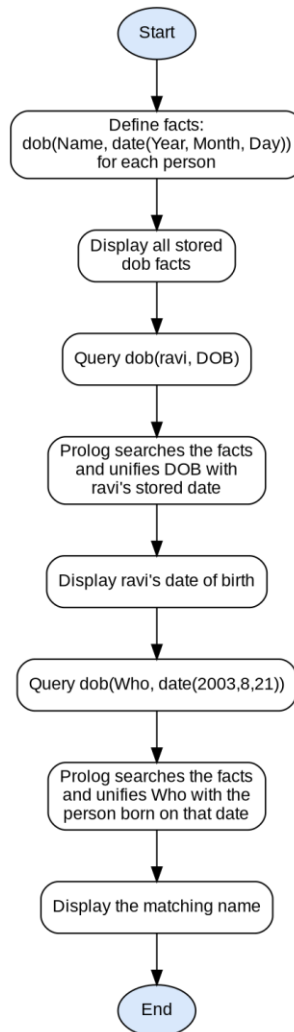
Aim

To write a Prolog program that stores a small database of people's names and dates of birth as facts, and answers queries against it.

Algorithm

1. Start.
2. Define facts of the form `dob(Name, date(Year, Month, Day))` for each person in the database.
3. Display all the stored dob facts.
4. Query `dob(ravi, DOB)` to find a specific person's date of birth.
5. Prolog searches the facts and unifies DOB with ravi's stored date.
6. Display the result of that query.
7. Query `dob(Who, date(2003, 8, 21))` to find the person born on a specific date.
8. Prolog searches the facts and unifies Who with the matching person.
9. Display the result of that query.
10. Stop.

Flowchart



Program

```

dob(ravi, date(2003, 5, 14)).
dob(anitha, date(2002, 11, 2)).
dob(suresh, date(2004, 1, 30)).
dob(kavya, date(2003, 8, 21)).

```

main :-

```

    format("Name-DOB Database~n"),
    format("-----~n"),
    forall(dob(Name, date(Y, M, D)),
        format("~w~t~15|: ~w-~w-~w~n", [Name, D, M, Y])),
    format("~nQuery: dob(ravi, DOB)~n"),
    dob(ravi, date(Y1, M1, D1)),

```

```

    format("Result: ravi was born on ~w-~w-~w~n", [D1, M1,
Y1]),
    format("~nQuery: dob(Who, date(2003, 8, 21))~n"),
    dob(Who, date(2003, 8, 21)),
    format("Result: ~w was born on 21-8-2003~n", [Who]).

```

Result

The program was executed successfully and the output was verified as correct.

?- main.

Name-DOB Database

```

-----
ravi           : 14-5-2003
anitha         : 2-11-2002
suresh         : 30-1-2004
kavya          : 21-8-2003

```

Query: dob(ravi, DOB)

Result: ravi was born on 14-5-2003

Query: dob(Who, date(2003, 8, 21))

Result: kavya was born on 21-8-2003

true.

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal:

The screenshot shows a Visual Studio Code window titled "name_dob.pl - AI-Lab - Visual Studio Code". The Explorer sidebar on the left lists files under "AI-Lab": sum_n.pl, name_dob.pl (selected), student_teacher.pl, planets.pl, and hanoi.pl. The main editor displays the Prolog code in name_dob.pl:

```

1  dob(ravi, date(2003, 5, 14)).
2  dob(anitha, date(2002, 11, 2)).
3  dob(suresh, date(2004, 1, 30)).
4  dob(kavya, date(2003, 8, 21)).
5
6  main :-
7      format("Name-DOB Database~n"),
8      format("-----~n"),
9      forall(dob(Name, date(Y, M, D)),
10         format("~w-t-15|: ~w-~w-~w~n", [Name, D, M, Y])),
11         format("~nQuery: dob(ravi, DOB)-~n"),
12         dob(ravi, date(Y1, M1, D1)),
13         format("Result: ravi was born on ~w-~w-~w~n", [D1, M1, Y1]),
14         format("~nQuery: dob(who, date(2003, 8, 21))-~n"),
15         dob(who, date(2003, 8, 21)),
16         format("Result: ~w was born on 21-8-2003~n", [who])).

```

The TERMINAL tab shows the execution of the Prolog code:

```

PS C:/Users/Athavan/AI-Lab> swipl name_dob.pl
Welcome to SWI-Prolog (threaded, 64 bits, version 9.0.4)

?- main.
Name-DOB Database
-----
ravi      : 14-5-2003
anitha    : 2-11-2002
suresh    : 30-1-2004
kavya     : 21-8-2003
Query: dob(ravi, DOB)
Result: ravi was born on 14-5-2003
Query: dob(who, date(2003, 8, 21))
Result: kavya was born on 21-8-2003
true.
?-

```

The status bar at the bottom indicates the current position is "Ln 1, Col 1" and the encoding is "Prolog UTF-8 LF Spaces: 4".